Huawei Access Node KR-ACC-12 Hardware/Environmental Fault Resolution SOP

Node: KR-ACC-12 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6267

Description

The Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Huawei Access Node KR-ACC-12 at KR Puram Access is a complex issue that requires a thorough understanding of the underlying protocol and hardware mechanics. The issue may be caused by a combination of factors including hardware failures, power supply redundancy loss, or environmental factors that affect the node's performance. In this SOP, we will focus on the resolution of ASIC memory parity errors and line card crashes, which are the most common causes of this failure domain. The Huawei Access Node KR-ACC-12 uses a distributed architecture, where each line card is responsible for processing and forwarding data packets. The ASIC memory parity errors occur when the line card's memory fails to maintain parity, causing data corruption and eventually leading to a system crash. The power supply redundancy loss occurs when one of the redundant power supplies fails, causing a loss of power to the node. To resolve this issue, we need to first identify the root cause of the problem, which may involve running diagnostic commands to isolate the faulty line card or power supply. Once the root cause is identified, we can proceed with the resolution steps, which may involve replacing the faulty line card or power supply, or performing a system restart to recover from the failure.

Pre-requisite Checks

- Verify that the node is in a stable state and that all alarms have been cleared.
- Check the redundant link status to ensure that it is functioning properly.
- Dispatch remote hands to KR Puram Access to physically access the node, if necessary.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	Medium
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display board info, Expected Output: Board ID: KR-ACC-12, Board Type: Huawei Access Node, Description: This command is used to display the board information of the node, including the board ID and type. This helps to identify the specific node in question.

Command: display board diag, Expected Output: Board Diag Result: Pass, Description: This command is used to display the diagnostic result of
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the board, which helps to identify any hardware or software issues.

Command: display env info, Expected Output: Temperature: 25°C,
Humidity: 60%, Description: This command is used to display the
environmental information of the node, including the temperature and
humidity levels. This helps to identify any environmental factors that
may be affecting the node's performance.
```

Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace linecard KR-ACC-12, Description: This step involves replacing the faulty line card with a new one. The command is used to remove the faulty line card and install a new one.

Step 2: Step: Restart the system, Command: system restart, Description: This step involves restarting the system to recover from the failure. The command is used to restart the system and bring it back to a stable state.

Post-Check Verification

```
Command: display board info, Expected Output: Board ID: KR-ACC-12,
Board Type: Huawei Access Node, Description: This command is used to
verify that the board information is correct after the resolution
steps have been completed.

Command: display board diag, Expected Output: Board Diag Result: Pass,
Description: This command is used to verify that the diagnostic result
of the board is correct after the resolution steps have been
completed.
```

Roll-back Procedures

- Step: Rollback the system configuration, Command: system rollback, Description: This step involves rolling back the system configuration to a previous state in case the fix fails.

- Step: Restore the original configuration, Command: system restore, Description: This step involves restoring the original configuration of the system in case the fix fails.